3 INTEGERS

Figure 3.1 The peak of Mount Everest. (credit: Gunther Hagleitner, Flickr)

Chapter Outline

Introduction

At over 29,000 feet, Mount Everest stands as the tallest peak on land. Located along the border of Nepal and China, Mount Everest is also known for its extreme climate. Near the summit, temperatures never rise above freezing. Every year, climbers from around the world brave the extreme conditions in an effort to scale the tremendous height. Only some are successful. Describing the drastic change in elevation the climbers experience and the change in temperatures requires using numbers that extend both above and below zero. In this chapter, we will describe these kinds of numbers and operations using them.

3.1 Introduction to Integers

Learning Objectives

By the end of this section, you will be able to:

- Locate positive and negative numbers on the number line
- Order positive and negative numbers
- Find opposites
- Simplify expressions with absolute value
- Translate word phrases to expressions with integers

Be Prepared!

Before you get started, take this readiness quiz.

1. Plot $0, 1, \text{and } 3$ on a number line.
   If you missed this problem, review Example 1.1.
2. Fill in the appropriate symbol: $(=, <, \text{or } >)$: $2\_\_\_4$
   If you missed this problem, review Example 2.3.

Locate Positive and Negative Numbers on the Number Line

Do you live in a place that has very cold winters? Have you ever experienced a temperature below zero? If so, you are already familiar with negative numbers. A **negative number** is a number that is less than $0.$ Very cold temperatures are measured in degrees below zero and can be described by negative numbers. For example, $-1°\text{F}$ (read as "negative one degree Fahrenheit") is 1 degree below $0.$ A minus sign is shown before a number to indicate that it is negative. **Figure 3.2** shows $-20°\text{F},$ which is 20 degrees below $0.$

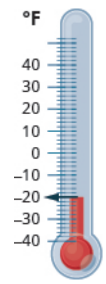


Figure 3.2 Temperatures below zero are described by negative numbers.

Temperatures are not the only negative numbers. A bank overdraft is another example of a negative number. If a person writes a check for more than he has in his account, his balance will be negative.

Elevations can also be represented by negative numbers. The elevation at sea level is $0\text{ feet}.$ Elevations above sea level are positive and elevations below sea level are negative. The elevation of the Dead Sea, which borders Israel and Jordan, is about $1{,}302\text{ feet}$ below sea level, so the elevation of the Dead Sea can be represented as $-1{,}302\text{ feet}.$ See **Figure 3.3**.

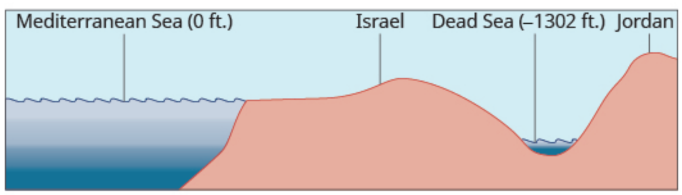


Figure 3.3 The surface of the Mediterranean Sea has an elevation of $0\text{ ft}.$ The diagram shows that nearby mountains have higher (positive) elevations whereas the Dead Sea has a lower (negative) elevation.

Depths below the ocean surface are also described by negative numbers. A submarine, for example, might descend to a depth of $500\text{ feet}.$ Its position would then be -500 feet as labeled in **Figure 3.4**.

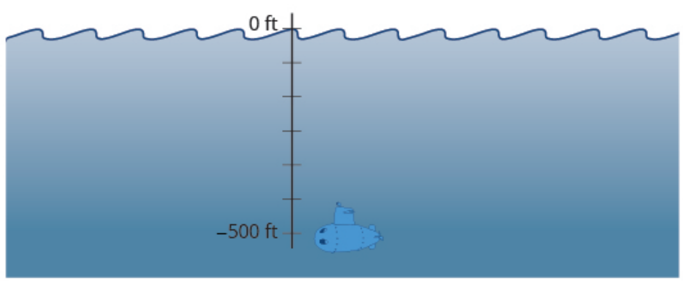


Figure 3.4 Depths below sea level are described by negative numbers. A submarine 500 ft below sea level is at $-500\text{ ft}.$

Both positive and negative numbers can be represented on a number line. Recall that the number line created in **Add Whole Numbers** started at 0 and showed the counting numbers increasing to the right as shown in **Figure 3.5**. The counting numbers $(1, 2, 3, \ldots)$ on the number line are all positive. We could write a plus sign, $+,$ before a positive number such as $+2$ or $+3,$ but it is customary to omit the plus sign and write only the number. If there is no sign, the number is assumed to be positive.

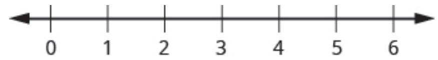


Figure 3.5

Now we need to extend the number line to include negative numbers. We mark several units to the left of zero, keeping the intervals the same width as those on the positive side. We label the marks with negative numbers, starting with -1 at the first mark to the left of 0, -2 at the next mark, and so on. See **Figure 3.6**.

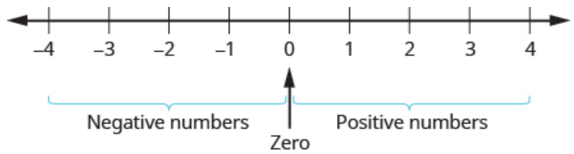


Figure 3.6 On a number line, positive numbers are to the right of zero. Negative numbers are to the left of zero. What about zero? Zero is neither positive nor negative.

The arrows at either end of the line indicate that the number line extends forever in each direction. There is no greatest positive number and there is no smallest negative number.

MANIPULATIVE MATHEMATICS

Doing the Manipulative Mathematics activity "Number Line-part 2" will help you develop a better understanding of integers.

EXAMPLE 3.1

Plot the numbers on a number line:

ⓐ 3 ⓑ -3 ⓒ -2

Solution

Draw a number line. Mark 0 in the center and label several units to the left and right.

ⓐ To plot $3,$ start at 0 and count three units to the right. Place a point as shown in **Figure 3.7**.

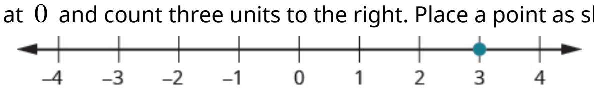


Figure 3.7

ⓑ To plot $-3,$ start at 0 and count three units to the left. Place a point as shown in **Figure 3.8**.

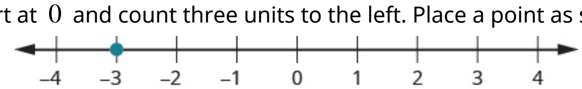


Figure 3.8

ⓒ To plot $-2,$ start at 0 and count two units to the left. Place a point as shown in **Figure 3.9**.

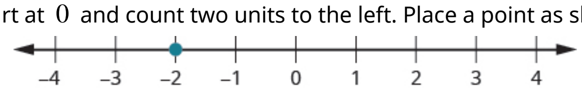


Figure 3.9

TRY IT : : 3.1 Plot the numbers on a number line.

ⓐ 1 ⓑ -1 ⓒ -4

TRY IT : : 3.2 Plot the numbers on a number line.

ⓐ -4 ⓑ 4 ⓐ -1

Order Positive and Negative Numbers

We can use the number line to compare and order positive and negative numbers. Going from left to right, numbers increase in value. Going from right to left, numbers decrease in value. See **Figure 3.10**.

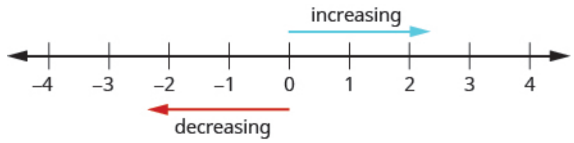


Figure 3.10

Just as we did with positive numbers, we can use inequality symbols to show the ordering of positive and negative numbers. Remember that we use the notation $a < b$ (read a *is less than* b) when a is to the left of b on the number line. We write $a > b$ (read a *is greater than* b) when a is to the right of b on the number line. This is shown for the numbers 3 and 5 in **Figure 3.11**.

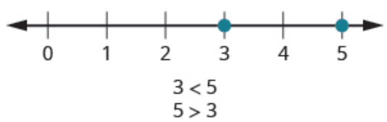
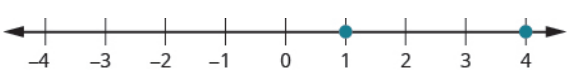


Figure 3.11 The number 3 is to the left of 5 on the number line. So 3 is less than $5,$ and 5 is greater than $3.$

The numbers lines to follow show a few more examples.

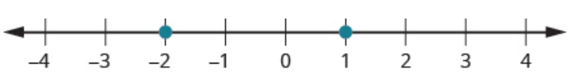
ⓐ



4 is to the right of 1 on the number line, so $4 > 1.$

1 is to the left of 4 on the number line, so $1 < 4.$

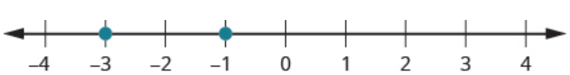
ⓑ



-2 is to the left of 1 on the number line, so $-2 < 1.$

1 is to the right of -2 on the number line, so $1 > -2.$

ⓒ



-1 is to the right of -3 on the number line, so $-1 > -3.$

-3 is to the left of -1 on the number line, so $-3 < -1.$

EXAMPLE 3.2

Order each of the following pairs of numbers using $<$ or $>$:

ⓐ $14\_\_\_6$ ⓑ $-1\_\_\_9$ ⓒ $-1\_\_\_-4$ ⓓ $2\_\_\_-20$

Solution

Begin by plotting the numbers on a number line as shown in Figure 3.12.

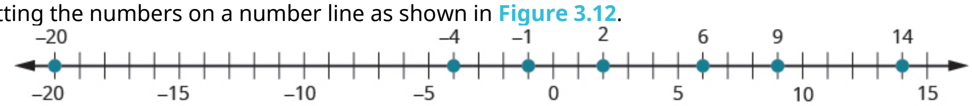


Figure 3.12

ⓐ Compare 14 and 6.	$14\_\_\_6$
14 is to the right of 6 on the number line.	$14 > 6$

ⓑ Compare −1 and 9.	$-1\_\_\_9$
−1 is to the left of 9 on the number line.	$-1 < 9$

ⓒ Compare −1 and −4.	$-1\_\_\_-4$
−1 is to the right of −4 on the number line.	$-1 > -4$

ⓓ Compare 2 and −20.	$-2\_\_\_-20$
2 is to the right of −20 on the number line.	$2 > -20$

TRY IT : : 3.3 Order each of the following pairs of numbers using $<$ or $>$.

ⓐ $15\_\_\_7$ ⓑ $-2\_\_\_5$ ⓒ $-3\_\_\_-7$ ⓓ $5\_\_\_-17$

TRY IT : : 3.4 Order each of the following pairs of numbers using $<$ or $>$.

ⓐ $8\_\_\_13$ ⓑ $3\_\_\_-4$ ⓒ $-5\_\_\_-2$ ⓓ $9\_\_\_-21$

Find Opposites

On the number line, the negative numbers are a mirror image of the positive numbers with zero in the middle. Because the numbers 2 and -2 are the same distance from zero, they are called **opposites**. The opposite of 2 is $-2,$ and the opposite of -2 is 2 as shown in Figure 3.13(a). Similarly, 3 and -3 are opposites as shown in Figure 3.13(b).

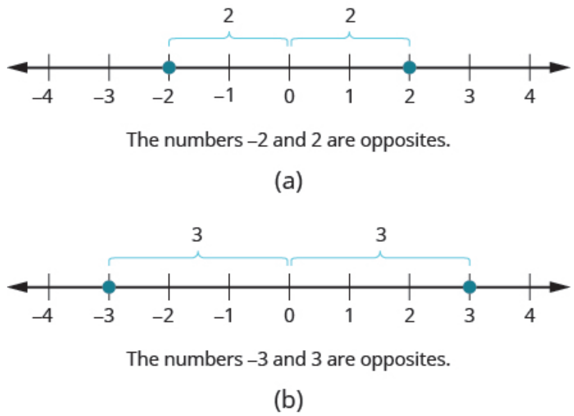


Figure 3.13

Opposite

The opposite of a number is the number that is the same distance from zero on the number line, but on the opposite side of zero.

EXAMPLE 3.3

Find the opposite of each number:

ⓐ 7 ⓑ -10

Solution

ⓐ The number -7 is the same distance from 0 as $7,$ but on the opposite side of $0.$ So -7 is the opposite of 7 as shown in **Figure 3.14**.

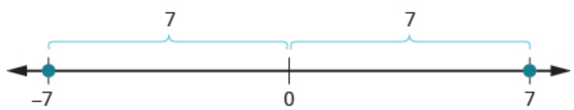


Figure 3.14

ⓑ The number 10 is the same distance from 0 as -10, but on the opposite side of $0.$ So 10 is the opposite of -10 as shown in **Figure 3.15**.

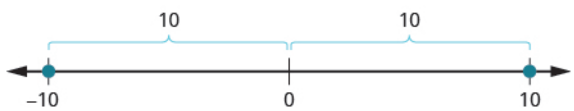


Figure 3.15

TRY IT : : 3.5 Find the opposite of each number:

ⓐ 4 ⓑ -3

TRY IT : : 3.6 Find the opposite of each number:

ⓐ 8 ⓑ -5

Opposite Notation

Just as the same word in English can have different meanings, the same symbol in algebra can have different meanings.

The specific meaning becomes clear by looking at how it is used. You have seen the symbol "$-$", in three different ways.

$10-4$	Between two numbers, the symbol indicates the operation of subtraction. We read $10-4$ as 10 *minus* 4.
-8	In front of a number, the symbol indicates a negative number. We read -8 as *negative eight*.
$-x$	In front of a variable or a number, it indicates the opposite. We read $-x$ as *the opposite of* x.
$-(-2)$	Here we have two signs. The sign in the parentheses indicates that the number is negative 2. The sign outside the parentheses indicates the opposite. We read $-(-2)$ as *the opposite of* -2.

Opposite Notation

$-a$ means the opposite of the number a

The notation $-a$ is read *the opposite of* a.

EXAMPLE 3.4

Simplify: $-(-6)$.

Solution

	$-(-6)$
The opposite of -6 is 6.	6

TRY IT : : 3.7 Simplify:

$-(-1)$

TRY IT : : 3.8 Simplify:

$-(-5)$

Integers

The set of counting numbers, their opposites, and 0 is the set of integers.

Integers

Integers are counting numbers, their opposites, and zero.

$$\ldots -3,\ -2,\ -1,\ 0,\ 1,\ 2,\ 3\ldots$$

We must be very careful with the signs when evaluating the opposite of a variable.

EXAMPLE 3.5

Evaluate $-x$:

ⓐ when $x=8$ ⓑ when $x=-8$.

Solution

ⓐ To evaluate $-x$ when $x = 8$, substitute 8 for x.

	$-x$
Substitute 8 for *x*.	$-(8)$
Simplify.	-8

ⓑ To evaluate $-x$ when $x = -8$, substitute -8 for x.

	$-x$
Substitute −8 for *x*.	$-(-8)$
Simplify.	8

TRY IT : : 3.9 Evaluate $-n$:

ⓐ when $n = 4$ ⓑ when $n = -4$

TRY IT : : 3.10 Evaluate: $-m$:

ⓐ when $m = 11$ ⓑ when $m = -11$

Simplify Expressions with Absolute Value

We saw that numbers such as 5 and -5 are opposites because they are the same distance from 0 on the number line. They are both five units from 0. The distance between 0 and any number on the number line is called the **absolute value** of that number. Because distance is never negative, the absolute value of any number is never negative.

The symbol for absolute value is two vertical lines on either side of a number. So the absolute value of 5 is written as $|5|$, and the absolute value of -5 is written as $|-5|$ as shown in Figure 3.16.

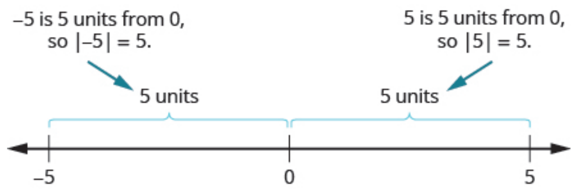


Figure 3.16

Absolute Value

The absolute value of a number is its distance from 0 on the number line.

The absolute value of a number n is written as $|n|$.

$$|n| \geq 0 \text{ for all numbers}$$

EXAMPLE 3.6

Simplify:

ⓐ $|3|$ ⓑ $|-44|$ ⓒ $|0|$

Solution

ⓐ

	$|3|$
3 is 3 units from zero.	3

ⓑ

	$|-44|$
−44 is 44 units from zero.	44

ⓒ

	$|0|$
0 is already at zero.	0

TRY IT : : 3.11 Simplify:

ⓐ $|12|$ ⓑ $-|-28|$

TRY IT : : 3.12 Simplify:

ⓐ $|9|$ ⓑ (b) $-|37|$

We treat absolute value bars just like we treat parentheses in the order of operations. We simplify the expression inside first.

EXAMPLE 3.7

Evaluate:

ⓐ $|x|$ when $x = -35$ ⓑ $|-y|$ when $y = -20$ ⓒ $-|u|$ when $u = 12$ ⓓ $-|p|$ when $p = -14$

Solution

ⓐ To find $|x|$ when $x = -35$:

	$|x|$
Substitute −35 for *x*.	$|-35|$
Take the absolute value.	35

ⓑ To find $|-y|$ when $y=-20$:

	$\mid -y\mid$
Substitute −20 for *y*.	$\mid-(-20)\mid$
Simplify.	$\mid 20\mid$
Take the absolute value.	20

ⓒ To find $-|u|$ when $u=12$:

	$-\mid u\mid$
Substitute 12 for *u*.	$-\mid 12\mid$
Take the absolute value.	-12

ⓓ To find $-|p|$ when $p=-14$:

	$-\mid p\mid$
Substitute −14 for *p*.	$-\mid-14\mid$
Take the absolute value.	-14

Notice that the result is negative only when there is a negative sign outside the absolute value symbol.

TRY IT : : 3.13

Evaluate:

ⓐ $|x|$ when $x=-17$ ⓑ $|-y|$ when $y=-39$ ⓒ $-|m|$ when $m=22$

ⓓ $-|p|$ when $p=-11$

TRY IT : : 3.14

ⓐ $|y|$ when $y=-23$ ⓑ $|-y|$ when $y=-21$ ⓒ $-|n|$ when $n=37$

ⓓ $-|q|$ when $q=-49$

EXAMPLE 3.8

Fill in $<, >,$ or $=$ for each of the following:

ⓐ $|-5|$___$-|-5|$ ⓑ 8___$-|-8|$ ⓒ -9___$-|-9|$ ⓓ $-|-7|$___-7

Solution

To compare two expressions, simplify each one first. Then compare.

ⓐ

	$\lvert -5 \rvert$ ___ $-\lvert -5 \rvert$
Simplify.	5 ___ -5
Order.	$5 > -5$

ⓑ

	8 ___ $-\lvert -8 \rvert$
Simplify.	8 ___ -8
Order.	$8 > -8$

ⓒ

	-9 ___ $-\lvert -9 \rvert$
Simplify.	-9 ___ -9
Order.	$-9 = -9$

ⓓ

	$-\lvert -7 \rvert$ ___ -7
Simplify.	-7 ___ -7
Order.	$-7 = -7$

TRY IT : : 3.15

Fill in $<, >,$ or $=$ for each of the following:

ⓐ $\lvert -9 \rvert$ ___ $-\lvert -9 \rvert$ ⓑ 2 ___ $-\lvert -2 \rvert$ ⓒ -8 ___ $\lvert -8 \rvert$ ⓓ $-\lvert -5 \rvert$ ___ -5

TRY IT : : 3.16

Fill in $<, >,$ or $=$ for each of the following:

ⓐ 7 ___ $-\lvert -7 \rvert$ ⓑ $-\lvert -11 \rvert$ ___ -11 ⓒ $\lvert -4 \rvert$ ___ $-\lvert -4 \rvert$ ⓓ -1 ___ $\lvert -1 \rvert$

Absolute value bars act like grouping symbols. First simplify inside the absolute value bars as much as possible. Then take the absolute value of the resulting number, and continue with any operations outside the absolute value symbols.

EXAMPLE 3.9

Simplify:

ⓐ $|9-3|$ ⓑ $4|-2|$

Solution

For each expression, follow the order of operations. Begin inside the absolute value symbols just as with parentheses.

ⓐ

	$|9-3|$
Simplify inside the absolute value sign.	$|6|$
Take the absolute value.	6

ⓑ

	$4|-2|$
Take the absolute value.	$4\cdot 2$
Multiply.	8

TRY IT : : 3.17 Simplify:

ⓐ $|12-9|$ ⓑ $3|-6|$

TRY IT : : 3.18 Simplify:

ⓐ $|27-16|$ ⓑ $9|-7|$

EXAMPLE 3.10

Simplify: $|8+7|-|5+6|$.

Solution

For each expression, follow the order of operations. Begin inside the absolute value symbols just as with parentheses.

	$|8+7|-|5+6|$
Simplify inside each absolute value sign.	$|15|-|11|$
Subtract.	4

TRY IT : : 3.19 Simplify: $|1+8|-|2+5|$

TRY IT : : 3.20 Simplify: $|9-5|-|7-6|$

EXAMPLE 3.11

Simplify: $24-|19-3(6-2)|$.

Solution

We use the order of operations. Remember to simplify grouping symbols first, so parentheses inside absolute value symbols would be first.

	$24 - \lvert 19 - 3(6 - 2) \rvert$
Simplify in the parentheses first.	$24 - \lvert 19 - 3(4) \rvert$
Multiply $3(4)$.	$24 - \lvert 19 - 12 \rvert$
Subtract inside the absolute value sign.	$24 - \lvert 7 \rvert$
Take the absolute value.	$24 - 7$
Subtract.	17

TRY IT : : 3.21 Simplify: $19 - \lvert 11 - 4(3 - 1) \rvert$

TRY IT : : 3.22 Simplify: $9 - \lvert 8 - 4(7 - 5) \rvert$

Translate Word Phrases into Expressions with Integers

Now we can translate word phrases into expressions with integers. Look for words that indicate a negative sign. For example, the word *negative* in “negative twenty” indicates -20. So does the word *opposite* in “the opposite of 20.”

EXAMPLE 3.12

Translate each phrase into an expression with integers:

ⓐ the opposite of positive fourteen ⓑ the opposite of -11 ⓒ negative sixteen

ⓓ two minus negative seven

Solution

ⓐ the opposite of fourteen
-14

ⓑ the opposite of −11
$-(-11)$

ⓒ negative sixteen
-16

ⓓ two minus negative seven
$2 - (-7)$

TRY IT : : 3.23

Translate each phrase into an expression with integers:

ⓐ the opposite of positive nine ⓑ the opposite of -15 ⓒ negative twenty

ⓓ eleven minus negative four

TRY IT : : 3.24

Translate each phrase into an expression with integers:

ⓐ the opposite of negative nineteen ⓑ the opposite of twenty-two ⓒ negative nine

ⓓ negative eight minus negative five

As we saw at the start of this section, negative numbers are needed to describe many real-world situations. We’ll look at some more applications of negative numbers in the next example.

EXAMPLE 3.13

Translate into an expression with integers:

ⓐ The temperature is 12 degrees Fahrenheit below zero. ⓑ The football team had a gain of 3 yards.

ⓒ The elevation of the Dead Sea is $1,302$ feet below sea level.

ⓓ A checking account is overdrawn by $\$40$.

Solution

Look for key phrases in each sentence. Then look for words that indicate negative signs. Don't forget to include units of measurement described in the sentence.

ⓐ	The temperature is 12 degrees Fahrenheit below zero.
Below zero tells us that 12 is a negative number.	$-12°F$

ⓑ	The football team had a gain of 3 yards.
A *gain* tells us that 3 is a positive number.	3 yards

ⓒ	The elevation of the Dead Sea is 1,302 feet below sea level.
Below sea level tells us that 1,302 is a negative number.	$-1,302$ feet

ⓓ	A checking account is overdrawn by $40.
Overdrawn tells us that 40 is a negative number.	$-\$40$

TRY IT : : 3.25 Translate into an expression with integers:

The football team had a gain of 5 yards.

TRY IT : : 3.26 Translate into an expression with integers:

The scuba diver was 30 feet below the surface of the water.

MEDIA : : ACCESS ADDITIONAL ONLINE RESOURCES

- Introduction to Integers (http://openstaxcollege.org/l/24introinteger)
- Simplifying the Opposites of Negative Integers (http://openstaxcollege.org/l/24neginteger)
- Comparing Absolute Value of Integers (http://openstaxcollege.org/l/24abvalue)
- Comparing Integers Using Inequalities (http://openstaxcollege.org/l/24usinginequal)

3.1 EXERCISES

Practice Makes Perfect

Locate Positive and Negative Numbers on the Number Line

In the following exercises, locate and label the given points on a number line.

1.
ⓐ 2 ⓑ -2
ⓒ -5

2.
ⓐ 5 ⓑ -5
ⓒ -2

3.
ⓐ -8 ⓑ 8
ⓒ -6

4.
ⓐ -7 ⓑ 7
ⓒ -1

Order Positive and Negative Numbers on the Number Line

In the following exercises, order each of the following pairs of numbers, using $<$ or $>$.

5.
ⓐ $9\_\_4$ ⓑ $-3\_\_6$
ⓒ $-8\_\_-2$ ⓓ $1\_\_-10$

6.
ⓐ $6\_\_2;$ ⓑ $-7\_\_4;$
ⓒ $-9\_\_-1;$ ⓓ $9\_\_-3$

7.
ⓐ $-5\_\_1;$ ⓑ $-4\_\_-9;$
ⓒ $6\_\_10;$ ⓓ $3\_\_-8$

8.
ⓐ $-7\_\_3;$
ⓑ $-10\_\_-5;$
ⓒ $2\_\_-6;$ ⓓ $8\_\_9$

Find Opposites

In the following exercises, find the opposite of each number.

9.
ⓐ 2 ⓑ -6

10.
ⓐ 9 ⓑ -4

11.
ⓐ -8 ⓑ 1

12.
ⓐ -2 ⓑ 6

In the following exercises, simplify.

13. $-(-4)$

14. $-(-8)$

15. $-(-15)$

16. $-(-11)$

In the following exercises, evaluate.

17. $-m$ when
ⓐ $m = 3$ ⓑ $m = -3$

18. $-p$ when
ⓐ $p = 6$ ⓑ $p = -6$

19. $-c$ when
ⓐ $c = 12$ ⓑ $c = -12$

20. $-d$ when
ⓐ $d = 21$
ⓑ $d = -21$

Simplify Expressions with Absolute Value

In the following exercises, simplify each absolute value expression.

21.
ⓐ $|7|$ ⓑ $|-25|$
ⓒ $|0|$

22.
ⓐ $|5|$ ⓑ $|20|$
ⓒ $|-19|$

23.
ⓐ $|-32|$ ⓑ $|-18|$
ⓒ $|16|$

24.
ⓐ $|-41|$ ⓑ $|-40|$
ⓒ $|22|$

In the following exercises, evaluate each absolute value expression.

25.
ⓐ $|x|$ when $x = -28$
ⓑ $|-u|$ when $u = -15$

26.
ⓐ $|y|$ when $y = -37$
ⓑ $|-z|$ when $z = -24$

27.
ⓐ $-|p|$ when $p = 19$
ⓑ $-|q|$ when $q = -33$

28.
ⓐ $-|a|$ when $a = 60$
ⓑ $-|b|$ when $b = -12$

In the following exercises, fill in $<$, $>$, $or =$ to compare each expression.

29.
ⓐ $-6\_\_|-6|$
ⓑ $-|-3|\_\_-3$

30.
ⓐ $-8\_\_|-8|$
ⓑ $-|-2|\_\_-2$

31.
ⓐ $|-3|\_\_-|-3|$
ⓑ $4\_\_-|-4|$

32.
ⓐ $|-5|\_\_-|-5|$
ⓑ $9\_\_-|-9|$

In the following exercises, simplify each expression.

33. $|8-4|$

34. $|9-6|$

35. $8|-7|$

36. $5|-5|$

37. $|15-7|-|14-6|$

38. $|17-8|-|13-4|$

39. $18-|2(8-3)|$

40. $15-|3(8-5)|$

41. $8(14-2|-2|)$

42. $6(13-4|-2|)$

Translate Word Phrases into Expressions with Integers

Translate each phrase into an expression with integers. Do not simplify.

43.
ⓐ the opposite of 8
ⓑ the opposite of -6
ⓒ negative three
ⓓ 4 minus negative 3

44.
ⓐ the opposite of 11
ⓑ the opposite of -4
ⓒ negative nine
ⓓ 8 minus negative 2

45.
ⓐ the opposite of 20
ⓑ the opposite of -5
ⓒ negative twelve
ⓓ 18 minus negative 7

46.
ⓐ the opposite of 15
ⓑ the opposite of -9
ⓒ negative sixty
ⓓ 12 minus 5

47. a temperature of 6 degrees below zero

48. a temperature of 14 degrees below zero

49. an elevation of 40 feet below sea level

50. an elevation of 65 feet below sea level

51. a football play loss of 12 yards

52. a football play gain of 4 yards

53. a stock gain of $\$3$

54. a stock loss of $\$5$

55. a golf score one above par

56. a golf score of 3 below par

Everyday Math

57. Elevation The highest elevation in the United States is Mount McKinley, Alaska, at $20,320$ feet above sea level. The lowest elevation is Death Valley, California, at 282 feet below sea level. Use integers to write the elevation of:
ⓐ Mount McKinley ⓑ Death Valley

58. Extreme temperatures The highest recorded temperature on Earth is $58°$ Celsius, recorded in the Sahara Desert in 1922. The lowest recorded temperature is $90°$ below $0°$ Celsius, recorded in Antarctica in 1983. Use integers to write the:
ⓐ highest recorded temperature
ⓑ lowest recorded temperature

59. State budgets In June, 2011, the state of Pennsylvania estimated it would have a budget surplus of $\$540$ million. That same month, Texas estimated it would have a budget deficit of $\$27$ billion. Use integers to write the budget:
ⓐ surplus ⓑ deficit

60. College enrollments Across the United States, community college enrollment grew by $1,400,000$ students from 2007 to 2010. In California, community college enrollment declined by $110,171$ students from 2009 to 2010. Use integers to write the change in enrollment:
ⓐ growth ⓑ decline

Writing Exercises

61. Give an example of a negative number from your life experience.

62. What are the three uses of the "−" sign in algebra? Explain how they differ.

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can…	Confidently	With some help	No-I don't get it!
locate positive and negative numbers on the number line.			
order positive and negative numbers.			
find opposites.			
simplify expressions with absolute value.			
translate word phrases to expressions with integers.			

ⓑ If most of your checks were:

…confidently. Congratulations! You have achieved the objectives in this section. Reflect on the study skills you used so that you can continue to use them. What did you do to become confident of your ability to do these things? Be specific.

…with some help. This must be addressed quickly because topics you do not master become potholes in your road to success. In math, every topic builds upon previous work. It is important to make sure you have a strong foundation before you move on. Who can you ask for help? Your fellow classmates and instructor are good resources. Is there a place on campus where math tutors are available? Can your study skills be improved?

…no—I don't get it! This is a warning sign and you must not ignore it. You should get help right away or you will quickly be overwhelmed. See your instructor as soon as you can to discuss your situation. Together you can come up with a plan to get you the help you need.